PHARMACOLOGY

GI Infections & Viral Hepatitis Therapy

SGL 3
Dr. Jangi Salai
GIT BLOCK

Pharmacology GIT Block · MEDucation by Mattin Majid

BEFORE YOU READ — ANSWER FROM MEMORY

1. When are antibiotics actually indicated for a bacterial GI infection?
2. Which drug class is used first-line for **both** HBV and HCV, and which class is HBV-only?
3. What is the very first step in treating C. difficile colitis — before any drug is started?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

*This lecture covers two linked but separate jobs: treating **GI infections** (mostly supportive/antibiotic) and treating **chronic viral hepatitis** (antivirals). They're kept in one file here since they were taught together — worth knowing in case you'd rather split it into two shorter PDFs.*

1 GI infections — the big picture

Cause	Approach
Viral	Rotavirus, norovirus (infants), astrovirus, enteric adenovirus — most common cause overall. **No antiviral drug** — conservative only: antipyretics, antiemetics, ORS.
Bacterial	**Antibiotics NOT recommended** unless infection causes bacteremia/septicemia or occurs in a debilitated patient. Most GI infections are self-limited (days) except in newborns, immunocompromised, or elderly, where they can be serious.

Dehydration is the main danger — **oral rehydration therapy (ORT)** matters more than antimicrobials for most cases. WHO low-osmolarity ORS (per litre): glucose 13.5g, NaCl 2.6g, trisodium citrate 2.9g, KCl 1.5g. Reduces stool output, vomiting, and need for IV fluids vs the original formula.

2 Bacterial GI infections — organism → antibiotic

Organism	First-line therapy	Safety note
Non-typhoid *Salmonella*	Ciprofloxacin 500mg/12h × 5–7d	Cipro contraindicated <14y, breastfeeding, pregnancy — use **Ceftriaxone** 1g IV/24h × 5d instead (safe in all three)
***E. coli* (ETEC/EPEC) — traveler's diarrhea**	Ciprofloxacin or Azithromycin 500mg/24h × 1–3d; Rifaximin 200mg/8h × 3d	Azithromycin and Rifaximin are safe <14y, breastfeeding, pregnancy
***Shigella* (bacillary dysentery)**	Ciprofloxacin 500mg/12h × 1–3d (WHO 1st line)	2nd line: Ceftriaxone 1g IV/24h or Azithromycin 500mg/24h × 1–3d
Campylobacter jejuni	Azithromycin 500mg/24h × 3d (WHO)	—
Vibrio cholerae	Doxycycline 300mg single dose (>8y); or Ciprofloxacin 500mg/12h × 3d; or Azithromycin 500mg/24h × 3d	Azithromycin is the appropriate alternative in pregnancy and children
***C. difficile* colitis (antibiotic-associated)**	**Stop the causative antibiotic first.** Oral metronidazole 250mg/6h (1st choice); if it fails → oral vancomycin 125mg/6h	Common culprits: ampicillin, amoxicillin, cephalosporins, clindamycin

The pattern: Azithromycin and Ceftriaxone are the pregnancy/paediatric-safe choices across almost every organism; Ciprofloxacin is avoided in children <14y, pregnancy, and breastfeeding unless there's no alternative.

Complicated intra-abdominal GI infections

Condition	Antibacterial options
Cholecystitis	

	Meropenem / imipenem-cilastatin · piperacillin-tazobactam · tigecycline · cefazolin/cefepime · ciprofloxacin/levofloxacin (each of the last three **plus metronidazole**)
Liver abscess & peritonitis	Piperacillin-tazobactam · tigecycline · meropenem

3 Viral hepatitis — treatment goals

Main goal for chronic HBV/HCV: prevent disease progression to **cirrhosis and hepatocellular carcinoma**. Additional goals: prevent mother-to-child transmission, manage HBV reactivation, treat extrahepatic manifestations. Diagnosis rests on serology (HBsAg, HBeAg, HBcAb, HCV-Ab etc.) ± PCR.

4 Antiviral drug classes for hepatitis

Class	Examples	Mechanism / key pharmacology	Key ADR / caution
Interferon-alpha (pegylated)	PEG-IFNα2a, PEG-IFNα2b	Antiviral (blocks penetration/uncoating/ assembly) + immunomodulatory + antitumour. SC/IM. Pegylation ↑bioavailability & half-life (α2b 40h, α2a 65h) → once-weekly instead of 3×/week. Effective for **both HBV and HCV**	Flu-like symptoms, bone marrow suppression, severe fatigue, depression (caution in psychiatric/epilepsy/thyroid disease), autoimmune thyroiditis
Nucleoside analogues	Lamivudine, Entecavir, Telbivudine	Phosphorylated to active triphosphate → inhibit HBV reverse transcriptase/DNA polymerase. **HBV only.** Lamivudine: PO once daily, bioavailability 82%, crosses placenta freely	High rates of resistant mutations; lactic acidosis, headache, dizziness; dose ↓ in renal (not hepatic) impairment
Nucleotide analogues	Tenofovir, Adefovir	Inhibit viral reverse transcriptase. **HBV only.** Tenofovir once daily — low resistance, active against lamivudine/ entecavir-resistant strains	Nausea, bloating
Polymerase inhibitor (nucleotide, HCV)	Sofosbuvir	Prodrug, triphosphorylated *within hepatocytes* during first-pass metabolism — activates right at its site of action, the liver. Once daily PO; combined with peg-IFN + ribavirin, or interferon-free	Small ↓Hb, headache, insomnia, fatigue, nausea; no dose adjustment needed for hepatic or mild–moderate renal impairment; no significant interaction with immunosuppressants or anti-HIV drugs
Ribavirin	—	Guanosine analogue; inhibits viral RNA-dependent RNA polymerase (also active vs influenza, RSV, HIV-1). Oral bioavailability 64% (↑ with fatty meal, ↓ with antacids). Given with SC peg-IFNα2b for chronic HCV	**Dose-dependent hemolytic anemia**; depression, fatigue, rash, insomnia, nausea. **Contraindicated:** anemia, end-stage renal failure, severe heart disease, pregnancy
Protease inhibitors	Telaprevir, Boceprevir, Atazanavir	Block the viral protease. Used with peg-IFNα + ribavirin to improve response and reduce resistance. Take with food to improve absorption	Telaprevir: anemia, rash, lipodystrophy, liver toxicity, diabetes. Atazanavir: no effect on insulin/lipids but ↑risk of proximal tubulopathy
Fixed-dose combo	Vosevi (sofosbuvir + velpatasvir + voxilaprevir)	Blocks 2 sites on viral RNA polymerase (sofosbuvir, velpatasvir) + the protease (voxilaprevir) — 3 proteins essential for HCV replication. For HCV without cirrhosis or with mild cirrhosis	Headache, nausea, diarrhea. **Must not combine with:** rosuvastatin, dabigatran, contraceptives, rifampicin, carbamazepine, phenobarbital, phenytoin (enzyme inducers cut its efficacy)

The split: Interferon-α works for **both** HBV and HCV. Nucleos(t)ide analogues (lamivudine, entecavir, telbivudine, tenofovir, adefovir) are **HBV reverse-transcriptase drugs only**. Sofosbuvir, ribavirin, and the protease inhibitors are the **HCV-specific** arm.

P Must-remember pearls

- Most GI infections are self-limited — **ORS**, not antibiotics, is the mainstay; antibiotics only for bacteremia/septicemia or debilitated patients.
- Azithromycin and Ceftriaxone are the pregnancy/paediatric-safe antibiotic choices across almost every GI organism; Ciprofloxacin usually isn't.
- *C. difficile* colitis = **stop the causative antibiotic first**, then metronidazole (1st line) or vancomycin (if it fails).
- Interferon-α treats **both** HBV and HCV; nucleos(t)ide analogues (lamivudine, entecavir, telbivudine, tenofovir, adefovir) are HBV-only reverse-transcriptase inhibitors.
- Sofosbuvir activates inside the liver itself during first-pass metabolism — that's why hepatic impairment doesn't need a dose adjustment.

Q Test yourself

1. Which antibiotic is the WHO-recommended first-line agent for Shigella?

A. Metronidazole
B. Ciprofloxacin
C. Vancomycin
D. Doxycycline

2. A pregnant woman develops V. cholerae infection. Which antibiotic is the safest choice?

A. Doxycycline
B. Ciprofloxacin
C. Azithromycin
D. Tetracycline

3. A patient develops watery diarrhea 5 days after starting clindamycin for a dental abscess. First step in management?

A. Start vancomycin immediately
B. Stop the clindamycin
C. Add loperamide
D. Start ciprofloxacin

4. Which hepatitis drug's activation depends on hepatic first-pass metabolism, converting it directly at its site of action?

A. Ribavirin
B. Telaprevir
C. Sofosbuvir
D. Interferon-alpha

5. Why is Vosevi contraindicated with carbamazepine?

A. Both cause hepatotoxicity
B. Carbamazepine is a CYP/enzyme inducer that reduces Vosevi's efficacy
C. Both lower seizure threshold
D. Carbamazepine binds Vosevi in the gut

ANSWERS — COVER UNTIL YOU'VE COMMITTED

1. — B — Ciprofloxacin 500mg/12h for 1–3 days is the WHO-recommended first line for Shigella.
2. — C — Azithromycin is the appropriate alternative for pregnant women and children with cholera.
3. — B — always stop the offending antibiotic first when C. difficile colitis is suspected.
4. — C — Sofosbuvir is a prodrug triphosphorylated in hepatocytes during first-pass metabolism, right at the liver.
5. — B — carbamazepine (like rifampicin, phenobarbital, phenytoin) induces metabolism and cuts Vosevi's efficacy.

SPACED REVIEW

☐ Day 1 | ☐ Day 3 | ☐ Day 7 | ☐ Day 21 | ☐ Pre-exam